1. Item 9: Understand the relationship between the context of the problem and representation of the mathematical solution...

(Understand how the real-world situation relates to its mathematical/graphical representation, and use this to interpret the feasibility and limitations of solutions)

- Highly relevant: The core challenge lies in understanding how changing viewpoint affects graphical representation.
  - The front view shows "door left, window right." From the back, left and right are reversed—as if viewing in a mirror.
  - Students must recognize that the drawings are not arbitrary but represent accurate projections from specific observer positions.
- This requires linking "3D object – 2D view – observer location," precisely what Item 9 emphasizes: understanding the correspondence between representation and real-world context.
- A student who ignores perspective and matches the rear view directly to the front will choose incorrectly—demonstrating that misinterpreting the representation leads to an invalid solution.

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2. Item 2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

(Judge whether a conclusion is reasonable within the real-world situation)

- Highly relevant: A student might initially select Option A (window on left, door on right) as the rear view. But if they reflect: "If this is the back, then the front would have a window on the left and door on the right—which contradicts the given front view"—they can eliminate the error.
- This kind of reasonableness check based on spatial logic is exactly the essence of Item 2: not accepting a superficial match, but asking, "Does this answer make sense for the actual structure of the garage?"
- The correct choice (Option D: rear view with window on left, door on right → front view with door on left, window on right) is valid precisely because it is spatially consistent.

3. Item 1: Interpret a mathematical result back into the real-world context

(Translate a mathematical/graphical result into meaningful real-world terms)

- Moderately relevant: Selecting the correct option implicitly means: "This rear view corresponds to a garage that, when viewed from the front, matches the given diagram."
- This judgment involves reinterpreting abstract symbols (lines, rectangles) as a real object's spatial configuration, embodying Item 1.
- Although no verbal explanation is required, correct selection inherently includes this act of "translation back to reality."

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4. Item 5: Interpret mathematical outcomes in a variety of formats... compare or evaluate two or more representations in relation to a situation

(Interpret results across multiple representations and assess their consistency within a context)

- Moderately relevant: Students must compare the front schematic with four candidate rear views.
- This involves coordinating two different visual representations of the same physical object and judging their consistency based on the shared context ("this is the same garage seen from opposite sides").
- Thus, the task assesses the ability to integrate and reconcile multiple graphical representations in a real-world task, as described in Item 5.

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5. Item 6: Use knowledge of how the real world impacts the outcomes... to make contextual judgments

(Apply real-world knowledge to inform interpretation or application of results)

- Weak but present relevance: Students draw on everyday experience: "When you look at a building from the back, left and right appear reversed compared to the front" (e.g., doors and windows swap sides).
- This commonsense understanding of spatial perspective supports correct application of the mirror-image rule.
- While not explicitly required, students lacking this real-world intuition may struggle with the viewpoint transformation.

1. 第9条：Understand the relationship between the context of the problem and representation of the mathematical solution...

（理解问题情境与数学/图形表征之间的关系，并据此解释解的可行性与局限）

- 高度相关：本题的核心挑战在于理解“视角转换”如何影响图形表征。
  - 正面图显示“左门右窗”，但从背面看时，左右互换（如同照镜子）。
  - 学生必须意识到：图纸不是随意绘制的，而是特定观察角度下的真实投影。
- 这要求学生建立“三维实体—二维视图—观察者位置”三者之间的联系，正是第9条强调的“理解表征与现实情境的对应关系”。
- 若学生忽略视角差异，直接按正面图匹配背面图，就会选错——这说明对表征的理解直接决定解的合理性。

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2. 第2条：Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

（判断数学结论在问题情境中是否合理）

- 高度相关：学生可能初步认为选项 A（左窗右门）正确，但若能反思：“如果这是背面，那正面就变成左窗右门了，与题干矛盾”，就能排除错误选项。
- 这种基于现实空间逻辑的合理性检验，正是第2条的核心——不盲目接受表面匹配，而是问：“这个答案在车库的真实结构中成立吗？”
- 正确解（应为 D：背面左窗右门 → 正面左门右窗）之所以成立，是因为它在空间逻辑上自洽。

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3. 第1条：Interpret a mathematical result back into the real world context

（将数学/图形结果还原到现实世界）

- 中等相关：选择正确选项后，学生实际上是在说：“这个背面图对应的车库，从前面看就是题干所示的样子。”
- 这一判断过程就是将抽象的图形符号（线条、方块）重新解释为真实物体的空间配置，体现了第1条的能力。
- 虽然题目未要求口头解释，但正确作答隐含了这种“回译”过程。

4. 第5条：Interpret mathematical outcomes in a variety of formats... compare or evaluate two or more representations in relation to a situation

（解读多种格式的数学结果，并在情境中比较或评估不同表征）

- 中等相关：学生需在正面示意图与四个背面视图之间进行跨视角比较。
- 这涉及处理同一对象的两种不同视觉表征，并基于情境（“这是同一个车库的不同视角”）判断其一致性。
- 因此，该题考查了在真实任务中协调多个图形表征的能力，符合第5条描述。

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5. 第6条：Use knowledge of how the real world impacts the outcomes... to make contextual judgments

（利用现实世界知识判断如何调整或应用结果）

- 弱相关但存在：学生需调用日常经验：“从房子后面看，左右是反的”（如从屋后看门窗位置与屋前相反）。
- 这种对现实空间关系的常识性理解，帮助他们正确应用镜像规则。
- 虽然未显性要求，但缺乏此常识的学生可能无法完成视角转换。